1 NAME OF THE MEDICINAL PRODUCT

TOBREX® (tobramycin 0.3%) Ophthalmic Solution 0.3%

2 QUALITATIVE AND QUANTITATIVE COMPOSITION

Active: tobramycin 0.3% (3 mg/mL)

Preservative: benzalkonium chloride 0.01% (0.1 mg/mL)

Excipients: See section 6.1

3 PHARMACEUTICAL FORM

Eye drops, solution

4 CLINICAL PARTICULARS

4.1 Therapeutic indications

TOBREX® Ophthalmic Solution is a topical antibiotic indicated in the treatment of external infections of the eye and its adnexa caused by susceptible bacteria. Appropriate monitoring of bacterial response to topical antibiotic therapy should accompany the use of TOBREX® Ophthalmic Solution. Clinical studies have shown tobramycin to be safe and effective for use in children.

4.2 Posology and method of administration

Adults and children older than 1 year:

- For mild and moderate infections, 1 to 2 drops instilled in the conjunctival sac with regular intervals with 4 hours between individual instillations for 7 days.

- For severe infections, 2 drops instilled every hour. After the condition has improved, dosage can be reduced.

Pediatric population

- Safety and effectiveness in pediatric patients below the age of 2 months has not been established.

Geriatric Use

- No overall clinical differences in safety or effectiveness have been observed between the elderly and other adult patients.

Renal Impairment

- The safety and efficacy of Tobrex Eye Drops in patients with renal impairment have not been established.

Hepatic Impairment

- The safety and efficacy of Tobrex Eye Drops in patients with hepatic impairment have not been established.

Method of administration

- For ocular use only.
- Keep the bottle tightly closed when not in use. After cap is removed, if tamper evident snap collar is loose, remove before using product.
- Nasolacrimal occlusion or gently closing the eyelid after administration is recommended. This may reduce the systemic absorption of medicinal products administered via the ocular route and result in a decrease in systemic adverse reactions.
- If more than one topical ophthalmic product is being used, the products must be administered at least 5 minutes apart. Eye ointments should be administered last.

- To prevent contamination of the dropper tip and solution, care must be taken not to touch the eyelids, surrounding areas or other surfaces with the dropper tip.

4.3 Contraindications

Hypersensitivity to the active substance or to any of the excipients.

4.4 Special warnings and precautions for use

- Sensitivity to topically administered aminoglycosides may occur in some patients. Severity of hypersensitivity reactions may vary from local effects to generalized reactions such as erythema, itching, urticarial, skin rash, anaphylaxis, anaphylactoid reactions, or bullous reactions. If hypersensitivity develops during use of this medicine, treatment should be discontinued.
- Cross-hypersensitivity to other aminoglycosides can occur, and the possibility that patients who become sensitized to topical ocular tobramycin may also be sensitive to other topical and/or systemic aminoglycosides should be considered.
- Serious adverse reactions including neurotoxicity, ototoxicity and nephrotoxicity have occurred in patients receiving systemic aminoglycoside therapy. Caution is advised when Tobrex Eye Drops are used concomitantly with systemic aminoglycosides.
- Caution should be exercised when prescribing Tobrex Eye Drops to patients with known or suspected neuromuscular disorders such as myasthenia gravis or Parkinson’s disease. Aminoglycosides may aggravate muscle weakness because of their potential effect on neuromuscular function.
- As with other antibiotic preparations, prolonged use of Tobrex Eye Drops may result in overgrowth of non-susceptible organisms, including fungi. If superinfection occurs, appropriate therapy should be initiated.
- Contact lens wear is not recommended during treatment of an ocular infection. Tobrex Eye Drops contains benzalkonium chloride which may cause eye irritation and is known to discolour soft contact lenses. Avoid contact with soft contact lenses. In case patients are

allowed to wear contact lenses, they must be instructed to remove contact lenses prior to application of this product and wait at least 15 minutes before reinsertion.

4.5 Interaction with other medicinal products and other forms of interaction

No clinically relevant interactions have been described with topical ocular dosing.

4.6 Pregnancy and lactation

Fertility

Studies have not been performed to evaluate the effect of topical ocular administration of Tobrex Eye Drops on human fertility.

Pregnancy

There are no or limited amount of data from the use of topical ocular tobramycin in pregnant women. Tobramycin does cross the placenta into the fetus after intravenous dosing in pregnant women. Tobramycin is not expected to cause ototoxicity from in utero exposure.

Studies in animals have shown reproductive toxicity at dosages considered sufficiently in excess of the maximal human dose derived from Tobrex Eye Drops so as to have limited clinical relevance. Tobramycin has not been shown to induce teratogenicity in rats or rabbits (See Section 5.3).

Tobrex Eye Drops should be used during pregnancy only if clearly needed.

Lactation

Tobramycin is excreted in human milk after systemic administration. It is unknown whether tobramycin is excreted in human milk following topical ocular administration. It is not likely that the amount of Tobramycin would be detectable in human milk or be capable of producing clinical effects in the infant following topical use of the product. However, a risk to the suckling child cannot be excluded. A decision must be made whether to discontinue breast-feeding or to discontinue/abstain from therapy taking into account the benefit of breast feeding for the child and the benefit of therapy for the woman.

4.7 Effects on ability to drive and use machines

Temporary blurred vision or other visual disturbances may affect the ability to drive or use machines. If blurred vision occurs at application, the patient must wait until the vision clears before driving or using machinery.

4.8 Undesirable effects

The following adverse reactions have been reported during clinical trials with Tobrex Eye Drops and are classified according to the subsequent convention: very common (≥1/10), common (≥1/100 to <1/10), uncommon (≥1/1,000 to <1/100), rare (≥1/10,000 to <1/1,000) and very rare (<1/10,000). Within each frequency grouping, adverse reactions are presented in order of decreasing seriousness.

System Organ Classification	**Adverse reactions**
Immune system disorders	*Uncommon*: hypersensitivity
Nervous system disorders	*Uncommon*: headache
Eye disorders	*Common*: ocular discomfort, ocular hyperaemia *Uncommon*: keratitis, corneal abrasion, visual impairment, vision blurred, erythema of eyelid, conjunctival oedema, eyelid oedema, eye pain, dry eye, eye discharge, eye pruritus, lacrimation increased
Skin and Subcutaneous Tissue Disorders	*Uncommon*: urticaria, dermatitis, madarosis, leukoderma, pruritus, dry skin

Additional adverse reactions identified from post-marketing surveillance include the following. Frequencies cannot be estimated from the available data.

System organ classification	**Adverse reactions**
Immune system disorders	anaphylactic reaction

Eye Disorder	eye allergy, eye irritation, eyelids pruritus
Skin and subcutaneous tissue disorders	Stevens-Johnson syndrome, erythema multiforme, rash

4.9 Overdose

Due to the characteristics of this preparation, no toxic effects are to be expected with an ocular overdose of this product, or in the event of accidental ingestion of the contents of one bottle.

5 PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: antibiotic ATC code: S01AA12

Mechanism of action

Tobramycin is a potent, broad-spectrum, fast-working bactericidal aminoglycoside antibiotic. It exerts its primary effect on bacterial cells by inhibiting polypeptide assembly and synthesis on the ribosome.

Mechanism of resistance

Resistance to tobramycin occurs by several different mechanisms including (1) alterations of the ribosomal subunit within the bacterial cell; (2) interference with the transport of tobramycin into the cell, and (3) inactivation of tobramycin by an array of adenylylating, phosphorylating, and acetylating enzymes. Genetic information for production of inactivating enzymes may be carried on the bacterial chromosome or on plasmids. Cross resistance to other aminoglycosides may occur.

Breakpoints

The breakpoints and the *in vitro* spectrum as mentioned below are based on systemic use. These breakpoints might not be applicable on topical ocular use of the medicinal product as higher concentrations are obtained locally and the local physical/chemical circumstances can influence the activity of the product on the site of administration. In accordance with EUCAST, the following breakpoints are defined for tobramycin:

- *Enterobacteriaceae S ≤2 mg/L, R >4 mg/L*
- *Pseudomonas spp.* S ≤4 mg/L, R >4 mg/L
- *Acinetobacter spp.* S ≤4 mg/L, R >4 mg/L
- *Staphylococcus spp.* S ≤1 mg/L, R >1 mg/L
- Not species-related S ≤2 mg/L, R >4 mg/L

Clinical efficacy against specific pathogens

The information listed below gives only an approximate guidance on probabilities whether microorganisms will be susceptible to tobramycin in this medicine. Bacterial species that have been recovered from external infections of the eye such as observed in conjunctivitis are presented here.

The prevalence of acquired resistance may vary geographically and with time for selected species and local information on resistance is desirable, particularly when treating severe infections. As necessary, expert advice should be sought when the local prevalence of resistance is such that the utility of tobramycin in at least some types of infections is questionable.

COMMONLY SUSCEPTIBLE SPECIES

- Aerobic Gram-positive microorganisms:
- Bacillus megaterium
- Bacillus pumilus
- Corynebacterium macginleyi
- Corynebacterium pseudodiphtheriticum
- Kocuria kristinae
- Staphylococcus aureus (methicillin susceptible – MSSA)
- Staphylococcus epidermidis (coagulase-positive and –negative)
- Staphylococcus haemolyticus (methicillin susceptible – MSSH)

- Streptococci (inlcuding some of the group A beta-hemolytic species, some nonhemo-lytic species, and some Streptococcus pneumoniae

Aerobic Gram-negative microorganisms:

- Acinetobacter calcoaceticus
- Acinetobacter junii
- Acinetobacter ursingii
- Citrobacter koseri
- Enterobacter aerogenes
- Escherichia coli
- H. aegyptius
- Haemophilus influenzae
- Klebsiella oxytoca
- Klebsiella pneumoniae
- Morganella morganii
- Moraxella catarrhalis
- Moraxella lacunata
- Moraxella oslonensis
- Some Neisseria species
- Proteus mirabilis
- Most Proteus vulgaris strains
- Pseudomonas aeruginosa
- Serratia liquifaciens

Anti-bacterial activity against other relevant pathogens

SPECIES FOR WHICH ACQUIRED RESISTANCE MIGHT BE A PROBLEM

- Acinetobacter baumanii
- Bacillus cereus
- Bacillus thuringiensis
- Kocuria rhizophila

- Staphylococcus aureus (methicillin resistant – MRSA)
- Staphylococcus haemolyticus (methicillin resistant –MRSH)
- Staphylococcus, other coagulase-negative spp.
- Serratia marcescens

INHERENTLY RESISTANT ORGANISMS

Aerobic Gram-positive microorganisms

- Enterococcus faecalis
- Streptococcus mitis
- Streptococcus pneumoniae
- Streptococcus sanguis
- Chryseobacterium indologenes

Aerobic Gram-negative microorganisms

- Haemophilus influenzae
- Stenotrophomonas maltophilia

Anaerobic Bacteria

- Propionibacterium acnes

Bacterial susceptibility studies demonstrate that in some cases, microorganisms resistant to gentamicin retain susceptibility to tobramycin.

PK/PD relationship

A specific PK/PD relationship has not been established for Tobrex. Published *in vitro* and *in vivo* studies have shown that tobramycin features a prolonged post-antibiotic effect, which effectively suppresses bacterial growth despite low serum concentrations.

Systemic administration studies have reported higher maximum concentrations with once daily compared to multiple daily dosing regimens. However, the weight of current evidence suggests that once daily systemic dosing is equally as efficacious as multiple-daily dosing. Tobramycin exhibits a concentration-dependent antimicrobial kill and greater efficacy with increasing levels of antibiotic above the MIC or minimum bactericidal concentration (MBC).

Data from clinical studies

Cumulative safety data from clinical studies are presented in Section 4.8.

Elderly population

No overall clinical differences in safety or efficacy have been observed between the elderly and other adult populations.

5.2 Pharmacokinetic properties

Absorption

Tobramycin is poorly absorbed across the cornea and conjunctiva with peak concentration of 3 micrograms/mL in aqueous humor after 2 hours followed by a rapid decline after topical administration of 0.3% tobramycin. Additionally, systemic absorption of tobramycin in human is poor after topical ocular administration of tobramycin. However, topical ocular tobramycin 0.3% delivers 527 ± 428 micrograms/mL tobramycin in human tears after a single dose. Ocular surface concentration generally exceeds the MIC of the most resistant isolates (MICs > 64 micrograms/mL).

Distribution

The systemic volume of distribution is 0.26 L/kg in man. Human plasma protein binding of tobramycin is low at less than 10%.

Biotransformation

Tobramycin is excreted in the urine primarily as unchanged drug.

Excretion

Tobramycin is excreted rapidly and extensively in the urine via glomerular filtration, primarily as unchanged drug. Systemic clearance was 1.43 ± 0.34 mL/min/kg for normal weight patients after intravenous administration and its systemic clearance decreased proportionally to renal function. The plasma half-life is approximately two hours.

Linearity/non-linearity pharmacokinetics

Ocular or systemic absorption with increasing dosing concentrations after topical ocular administration has not been evaluated. Therefore, the linearity of exposure with topical ocular dose could not be established.

Use in hepatic and renal impaired patients

Tobramycin pharmacokinetics with eye drops has not been studied in these patient populations.

Effect of age on pharmacokinetics

There is no change in tobramycin pharmacokinetics with older patients when compared to younger adults.

Use in pediatrics

Aminoglycosides including tobramycin has been commonly used among children, infants and neonates to treat serious Gram-negative infections. Tobrex (0.3% ophthalmic tobramycin) is approved for use in children. Clinical pharmacology of tobramycin in children has been described after systemic administration.

5.3 Preclinical safety data

Non-clinical data revealed no special hazard for humans from topical ocular exposure to tobramycin based on conventional repeated-dose topical ocular toxicity studies, genotoxicity or carcinogenicity studies. Effects in non-clinical reproductive and developmental studies with tobramycin were observed only at exposures considered sufficiently in excess of the maximum human ocular dosage indicating little relevance to clinical use.

6 PHARMACEUTICAL PARTICULARS

6.1 List of excipients

benzalkonium chloride

boric acid

tyloxapol

sodium chloride

sodium sulphate anhydrous

sulphuric acid and/or sodium hydroxide (to adjust pH)

purified water

6.2 Incompatibilities

Not applicable.

6.3 Special precautions for storage

Do not store above 30°C.

Keep out of reach and sight of children.

Do not use this medicine after the expiry date which is stated on the packaging.

Discard 4 weeks after first opening.

6.4 Nature and contents of container

Plastic container containing 5 mL.

6.5 Instructions for use and handling

No special requirements.

7 PRODUCT REGISTRATION HOLDER

Novartis Corporation (Malaysia) Sdn. Bhd.

Level 18, Imazium,

No. 8, Jalan SS21/37, Damansara Uptown,

47400 Petaling Jaya, Selangor.

Malaysian Package Insert

Information issued: 22 Mar 2019

Date of revision: 28 Oct 2024

® = registered trademark